Calibration and Frozen-Feature Adaptation for Shot Boundary Detection: A Reproducibility Study

Senh Manh Tang, Tien Manh Do, and Thai Son Tran

***Abstract*—Shot boundary detection (SBD) is a threshold-sensitive preprocessing step whose reported accuracy depends on model training, score calibration, temporal filtering, and operating-point selection. We present a reproducibility-focused study of frozen-feature adaptation for AutoShot. The study freezes the searched AutoShot representation, trains a binary cross-entropy head on cached 4864-dimensional features, and selects temperature scaling, Gaussian smoothing, and a decision threshold using five-fold validation only. The selected single-seed configuration achieves F1 scores of 0.8540 on SHOT, 0.9570 on BBC, and 0.7441 on ClipShots. Paired video bootstrap estimates improvements over the unsmoothed BCE control of 0.0162 on SHOT and 0.0457 on ClipShots, while the BBC interval includes zero. Temperature scaling reduces validation NLL from 0.013006 to 0.009591 and equal-width ECE from 0.007149 to 0.002214. However, class-balanced ECE remains 0.070405, showing that aggregate frame-level calibration metrics are optimistic under severe boundary sparsity. Controlled ablations further show that Gaussian smoothing changes boundary accuracy, whereas temperature scaling primarily improves probability quality. We release validation-selection manifests, per-video statistics, analysis code, and checksum-oriented artifact tooling; the reported training checkpoint remains limited to one seed.**

***Index Terms*—Shot boundary detection, short video, AutoShot, frozen features, temperature scaling, Gaussian smoothing, video understanding.**

I. Introduction

Shot boundary detection (SBD) partitions a video into temporally coherent shots by identifying frames where the camera view, scene content, or editing state changes. It is a common first step for video indexing, retrieval, summarization, and automated editing [1], [2]. Classical methods based on color histograms, edge changes, or motion cues are efficient, but they are fragile under illumination changes, fast camera motion, flashes, and modern editing effects.

Deep SBD models such as TransNet and TransNetV2 [5], [6] learn frame-level transition patterns directly from short frame windows. AutoShot [4] further targets short-video SBD by searching a specialized 3-D convolutional backbone for the SHOT dataset. This improves performance on short-form videos, but the neural architecture search (NAS) and subsequent retraining pipeline are costly to repeat. At the same time, short-video SBD remains difficult because boundaries are sparse, shots are short, and editing effects produce many hard negatives.

This paper asks two practical questions: how far can a searched AutoShot backbone be adapted without repeating NAS, and how much of the measured gain comes from probability calibration versus temporal filtering? We treat the problem as a reproducibility study rather than as a claim of a new backbone. The system reuses the 4864-dimensional AutoShot representation, trains only a binary classification head, and processes its frame scores with temperature scaling [10] and Gaussian smoothing.

The operating point is selected without test feedback. We construct five validation folds stratified by source dataset, fit temperature on the four training folds, and choose smoothing and threshold by mean macro-F1 across validation sources. This procedure independently recovers $T = 0.6618$, $\sigma = 2.0$, and $\theta = 0.10$. Test results are computed only after this manifest is frozen.

The central empirical result is that the simple BCE one-hot control is sufficient at training time. Focal Loss [9], many-hot supervision, and their combination change F1 only marginally. The selected post-processing configuration raises SHOT F1 from 0.8378 to 0.8540 and ClipShots F1 from 0.6983 to 0.7441. BBC remains effectively unchanged. Paired bootstrap analysis confirms positive boundary-level deltas on SHOT and ClipShots but not on BBC.

The main contributions are:

- We define a validation-only, source-stratified selection protocol that freezes temperature, smoothing, and threshold before test evaluation.
- We separate probability calibration from temporal filtering and show with protocol-matched ablations that Gaussian smoothing drives the boundary-level gain.
- We report paired video bootstrap intervals and calibration diagnostics that include adaptive and class-balanced ECE, exposing the effect of sparse positive frames.
- We provide strict cache identities, exact split/run manifests, a three-seed runner, and checksum-based artifact packaging. The currently reported checkpoint is explicitly identified as single-seed.

II. Related Work

Shot boundary detection. Shot boundary detection has long been an active research area in video processing. Early systems primarily measured frame-to-frame discontinuities using hand-crafted visual features. Methods relying on histogram differences, edge-change ratios, motion compensation, and thresholding-based heuristics are computationally simple. However, they are prone to mistaking lighting changes, camera motion, or object movement for actual shot boundaries [3]. Comprehensive survey studies categorize the major challenges

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in this domain into abrupt cuts, gradual transitions, illumination changes, object/camera motion, and dataset-dependent evaluation protocols [1], [2].

Deep SBD models. Following the success of deep learning in vision tasks, deep models were introduced to learn temporal patterns that are difficult to encode manually [13], [14]. TransNet models SBD as a dense frame-level prediction task over a sequence of low-resolution frames [6]. TransNetV2 further improves this concept by incorporating dilated 3-D convolutions, frame-similarity features, color histograms, and dual prediction heads for both single-frame and all-transition labels [5]. These models offer a fast and accurate inference pipeline, establishing them as strong baselines for practical SBD applications.

AutoShot and short-video SBD. Most existing datasets, such as BBC, RAI, and ClipShots, mainly represent documentaries, broadcast content, or longer user-generated videos. Recognizing this gap, AutoShot introduces the SHOT dataset specifically to capture short-video characteristics: brief duration, dense editing, vertical layouts, complex gradual transitions, and heavy special effects [4]. AutoShot leverages NAS [8], [11], [12] to search over approximately 83.9 million candidate architectures composed of 3-D separable convolution blocks and temporal Transformer blocks. While the searched backbone yields notable F1 improvements on SHOT, the computational cost of the search and retraining phases remains prohibitively high.

Imbalance, calibration, and temporal smoothing. Frame-level SBD is severely imbalanced because boundary frames are rare. Focal Loss reduces the contribution of easy examples and is therefore a plausible alternative to BCE [9]. SBD decisions are also threshold-sensitive, making score calibration and temporal stability important. Temperature scaling rescales logits without changing their ranking [10], while temporal smoothing suppresses isolated peaks. We evaluate these mechanisms separately rather than assuming that a more complex training loss is necessary.

III. Frozen-Feature Study Design

A. Problem Formulation

Given a video with T frames, SBD predicts a binary boundary sequence $\{\hat{b}_t\}_{t=1}^{T}$. A model first produces frame-level logits z_t. After calibration and smoothing, the final decision is

$$\hat{b}_t = \mathbb{1}[\tilde{p}_t > \theta]. \tag{1}$$

Consecutive positive predictions are grouped into one transition interval. Evaluation uses the canonical interval matcher with a tolerance of ± 2 frames.

B. Frozen AutoShot Representation

AutoShot [4] searches a short-video backbone composed of 3-D convolutional, similarity, histogram, and optional temporal blocks. Repeating this search is outside the intended adaptation budget. We freeze the searched backbone and cache its 4864-dimensional feature vector $h_t \in \mathbb{R}^{4864}$ for each frame. The selected A1 control trains only a compact decision head:

$$u_t = \mathrm{Dropout}(\mathrm{ReLU}(W_f h_t + c_f)), \qquad z_t = W_c u_t + c_c, \tag{2}$$

where $W_f \in \mathbb{R}^{1024\times 4864}$. The implementation retains a second output for controlled auxiliary-label ablations, but its loss weight is zero for A1 and the deployed B4 pipeline uses only the primary logit z_t.

The selected training objective is binary cross-entropy:

$$\mathcal{L}_{\mathrm{BCE}} = -y_t \log \sigma(z_t) - (1 - y_t)\log(1 - \sigma(z_t)). \tag{3}$$

This choice is empirical rather than assumed: the controlled experiments show no measurable benefit from replacing it with Focal Loss or adding many-hot auxiliary supervision.

C. Validation-Only Protocol Selection

Temperature scaling rescales the primary logit without changing its ranking:

$$p_t^{\mathrm{cal}} = \sigma\left(\frac{z_t}{T}\right). \tag{4}$$

Temperature scaling changes score calibration but cannot remove isolated temporal peaks. We therefore apply a one-dimensional Gaussian filter,

$$\tilde{p}_t = \sum_k G_\sigma(k) p_{t-k}^{\mathrm{cal}}, \tag{5}$$

We select the complete operating point using five-fold validation. Folds are stratified by source dataset. For each held-out fold, temperature is fitted on the other four folds by minimizing binary cross-entropy. We search $\sigma \in \{0, 1, 2, 3\}$ and $\theta \in \{0.05, 0.10, \ldots, 0.50\}$, scoring each pair by mean macro-F1 across validation sources and folds. Ties prefer lower smoothing and then the threshold nearest 0.10. After selecting σ and θ, the final temperature is fitted on all validation videos. This procedure selects $T = 0.6618$, $\sigma = 2.0$, and $\theta = 0.10$. The manifest containing folds, key hashes, search space, and selected values is written before test evaluation. We denote the resulting A1 plus temperature plus Gaussian configuration B4.

D. Cache and Run Provenance

The revised pipeline separates a fixed data seed from the training seed. Before feature extraction, it deterministically selects the training-video list and hashes that exact list. A sample cache is reusable only when its schema version, metadata hash, base-checkpoint hash, selected-key hash, video budget, sampling parameters, boundary width, and data seed all match. Each run writes the selected video IDs, per-video sample counts, skipped inputs, configuration, checkpoint hash, and logit-key hash. This strict identity replaces the earlier path-tolerant cache check that could not prove which 400 videos produced the historical cache.

E. Training-Time Alternatives

We evaluate Focal Loss and boundary-aware many-hot targets as alternatives, not as components of the selected method. For binary targets, the implemented Focal Loss is

$$\mathcal{L}_{\mathrm{focal}} = -\alpha_t (1 - p_t^*)^\gamma \log(p_t^*), \tag{6}$$

where $p_t^* = p_t$ for a positive target and $1 - p_t$ otherwise, and $\alpha_t = \alpha y_t + (1 - \alpha)(1 - y_t)$. The experiments use $\gamma = 1$ and

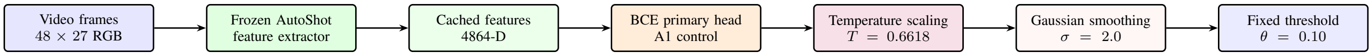

Fig. 1. Selected AutoShotV2 pipeline. B4 applies validation-selected temperature scaling and Gaussian smoothing to the controlled A1 BCE logits.

TABLE I
FIXED REPRODUCIBLE EVALUATION PROTOCOL

Item	Setting
Training objective	BCE, one-hot labels
Training / data seed	42 / 42
Training budget	20 epochs, batch size 512
Sampling	At most 160 frames/video, 3 negatives/positive
Validation size	200 videos
Validation selection	5 source-stratified folds
Selection criterion	Mean macro-F1 across sources/folds
Selected temperature	0.6618
Gaussian smoothing	$\sigma = 2.0$
Decision threshold	$\theta = 0.10$
Matching tolerance	± 2 frames
Bootstrap	10,000 video resamples, seed 42

$\alpha = 0.5$. The many-hot alternative expands each boundary label by one neighboring frame and applies an auxiliary loss with weight 0.3. These variants are retained in the ablation to establish that the B4 gain comes primarily from post-processing.

IV. EXPERIMENTS

A. Datasets and Fixed Protocol

We evaluate on the 200-video SHOT test split [4], all 11 BBC Planet Earth videos, and the 500-video ClipShots test split [7]. SHOT is the primary short-video benchmark, BBC tests long documentary footage, and ClipShots tests open-world transfer.

All headline numbers use the same controlled A1 logits and the B4 operating point frozen by the validation-only procedure in Section III. Precision, recall, and F1 are micro-averaged from transition-level TP/FP/FN counts. Confidence intervals resample paired videos rather than individual frames. They quantify test-set sampling uncertainty, not training-seed uncertainty.

B. Protocol-Matched Results

Table II compares configurations that share the controlled Phase 2 protocol. B4 is selected exclusively by the frozen five-fold validation manifest. Relative to A1, B4 improves SHOT by 1.62 F1 points and ClipShots by 4.57 points, while BBC is unchanged after rounding. Temperature scaling alone preserves the validation-selected operating point; Gaussian smoothing is the component that changes boundary decisions.

Table III reports the corresponding B4 operating points and video-level bootstrap intervals. BBC has only 11 sampling units, so its interval should be interpreted cautiously. ClipShots has high recall but lower precision, indicating cross-domain false positives rather than a general failure to detect transitions.

TABLE II
PROTOCOL-MATCHED F1 SCORES. B4 IS THE SELECTED METHOD

Configuration	SHOT	BBC	ClipShots
A1 – BCE + one-hot	0.8378	0.9570	0.6983
P1 – A1 + Gaussian	0.8432	0.9436	0.7519
P2 – A1 + temperature	0.8378	0.9570	0.6983
B4 – A1 + temperature + Gaussian	**0.8540**	**0.9570**	**0.7441**
B5 – Focal + many-hot + B4	0.8542	0.9551	0.7409

TABLE III
B4 OPERATING POINT AND VIDEO-LEVEL BOOTSTRAP 95% CONFIDENCE INTERVALS

Dataset	F1	F1 95% CI	Precision	Recall	Videos
SHOT	0.8540	[0.8329, 0.8733]	0.8617	0.8466	200
BBC	0.9570	[0.9310, 0.9717]	0.9733	0.9412	11
ClipShots	0.7441	[0.6997, 0.7841]	0.6498	0.8703	500

TABLE IV
PAIRED VIDEO BOOTSTRAP DIFFERENCE, B4 MINUS A1

Dataset	ΔF1	95% CI	Excl. 0
SHOT	0.0162	[0.0093, 0.0236]	Yes
BBC	-0.0000	[-0.0033, 0.0041]	No
ClipShots	0.0457	[0.0267, 0.0670]	Yes

TABLE V
LITERATURE-REPORTED F1 SCORES (CONTEXT ONLY)

Method	SHOT	BBC	ClipShots
AutoShot (reported)	0.8410	0.9710	0.7870
TransNetV2 (reported)	0.7993	0.9620	0.7760

Table IV directly bootstraps the per-video difference between A1 at its validation-selected threshold and B4 at its frozen operating point. The intervals support a positive B4 delta on SHOT and ClipShots. The BBC interval contains zero, so we make no improvement claim for that dataset.

For context only, Table V lists numbers copied from the original AutoShot and TransNetV2 reports. These values use their authors' protocols and are not mixed into the controlled selection above.

C. Calibration Diagnostics

We evaluate frame probabilities on the combined 200-video validation split using negative log-likelihood (NLL), Brier score, expected calibration error with 15 equal-width bins (ECE), equal-mass adaptive ECE (AECE), and class-balanced ECE (B-ECE). Table VI shows that temperature scaling improves all diagnostics. The much larger B-ECE reveals that aggregate frame metrics are dominated by negative

TABLE VI
FRAME-LEVEL VALIDATION CALIBRATION DIAGNOSTICS

Variant	NLL	Brier	ECE	AECE	B-ECE
Before temperature scaling	0.013006	0.002194	0.007149	0.007149	0.080506
After temperature scaling	0.009591	0.001911	0.002214	0.002243	0.070405

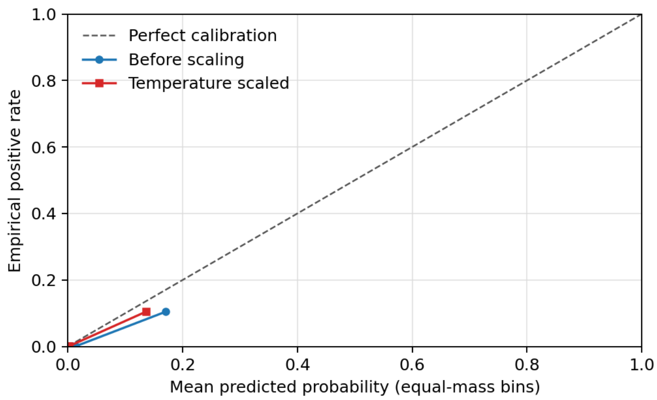

Fig. 2. Reliability diagram on the combined validation split. Temperature scaling reduces aggregate error. Equal-mass bins expose sparsely populated confidence regions.

TABLE VII
TEST CALIBRATION AT THE VALIDATION-SELECTED TEMPERATURE

Dataset	NLL	Brier	ECE	AECE	B-ECE
SHOT	0.021689	0.004808	0.003014	0.002844	0.133525
BBC	0.013305	0.001662	0.000851	0.001651	0.092471
ClipShots	0.007654	0.001565	0.002520	0.002521	0.156939

frames. Validation values remain diagnostic because the same split fits temperature.

Table VII reports untouched-test calibration using the temperature selected on validation. ClipShots has the largest B-ECE despite a small aggregate ECE, consistent with domain-shifted false positives.

D. Controlled Training Ablation

Table VIII expands the training-time comparison. A2, A3, and B1 show that Focal Loss and many-hot supervision do not improve the A1 control. B5 combines these training changes with B4 and is numerically similar on SHOT but slightly weaker on BBC and ClipShots. We therefore avoid attributing the result to the more complex training objective. A0 is omitted because its per-dataset best-threshold protocol is not a direct controlled comparison.

E. Efficiency and Reproducibility

The backbone remains frozen, so only the Phase 2 head is optimized. Table IX reports parameter and run-artifact statistics. The CPU post-processing measurement covers temperature scaling and Gaussian smoothing on cached logits. End-to-end video decoding and backbone inference are excluded. The original run artifact did not retain elapsed training time, so we do not estimate or reconstruct it.

TABLE VIII
CONTROLLED PHASE 2 ABLATION (A0 HISTORICAL BASELINE EXCLUDED)

Configuration	SHOT	BBC	ClipShots
A1 – BCE + one-hot	0.8378	0.9570	0.6983
A2 – Focal only	0.8378	0.9567	0.6967
A3 – Many-hot only	0.8375	0.9563	0.7005
P1 – Gaussian only	0.8432	0.9436	0.7519
P2 – Temperature only	0.8378	0.9570	0.6983
B1 – Focal + many-hot	0.8384	0.9559	0.7006
B4 – Temperature + Gaussian (selected)	**0.8540**	**0.9570**	**0.7441**
B5 – Full candidate, no EMA	0.8542	0.9551	0.7409

TABLE IX
EFFICIENCY AND RUN-ARTIFACT STATISTICS

Item	Value
Frozen backbone parameters	14,299,202
Trainable Phase 2 head parameters	4,983,810
Deployed primary-path parameters	4,982,785
Training videos / sampled frames	400 / 33,174
Training epochs / seed	20 / 42
Post-process CPU cost	21.1 ms / 10^6 frames

TABLE X
CLIPSHOTS RECALL BY GROUND-TRUTH TRANSITION TYPE

Type	GT	Matched	Missed	Recall
Cut	4907	4412	495	0.8991
Gradual	2302	1862	440	0.8089

F. ClipShots Transition-Type Recall

We regenerate the ClipShots breakdown directly from the B4 logits and `test.json`. One malformed annotation with reversed endpoints is normalized by sorting the pair. A ground-truth interval is a cut when its width is at most one frame and gradual otherwise. Predictions are class-agnostic, so they are matched once and recall is grouped by the matched ground-truth type. Type-specific precision and F1 are not reported because the model does not predict a transition type.

The gradual-transition recall of 0.8089 is lower than the cut recall of 0.8991, but it is not the dominant source of the overall ClipShots gap. The aggregate ClipShots precision is only 0.6498, indicating that open-world effects and domain shift create many false-positive peaks.

G. Archived Checkpoint Results and Limitations

For auditability, Table XI preserves the earlier deploy checkpoint numbers. Only result JSON files remain for this checkpoint; its test logits are absent. These numbers are therefore excluded from the headline and confidence-interval analysis.

The study has four important limitations. First, the reported head was trained with one seed; bootstrap intervals capture video sampling uncertainty but not optimization variability. The revised runner fixes data seed 42 and schedules training seeds 42, 43, and 44, but the original AutoShot base checkpoint needed to execute those runs is not included in the

TABLE XI
ARCHIVED DEPLOY CHECKPOINT (RESULT JSON ONLY)

Archived protocol	SHOT	BBC	ClipShots
Deploy checkpoint, fixed threshold	0.8545	0.9656	0.7530
Deploy checkpoint, best sweep	0.8545	0.9656	0.7557

present artifact bundle. Second, the historical sample cache recorded 400 videos and 33,174 frames but not the exact selected IDs; strict manifests fix this for future runs but cannot retroactively prove the old cache contents. Third, validation calibration diagnostics reuse the temperature-selection data. Fourth, the frozen representation does not fully control cross-domain false positives on ClipShots.

V. CONCLUSION AND DISCUSSION

We presented a reproducibility-focused study of calibration and frozen-feature adaptation for AutoShot. The selected configuration trains a BCE one-hot head on cached features and applies temperature scaling and Gaussian smoothing chosen by source-stratified, validation-only cross-validation. It reaches F1 scores of 0.8540 on SHOT, 0.9570 on BBC, and 0.7441 on ClipShots for the reported seed.

The ablation clarifies the source of the gain: Focal Loss and many-hot supervision do not improve the BCE control, whereas Gaussian smoothing changes boundary decisions and temperature scaling improves probability diagnostics. Paired bootstrap intervals support gains on SHOT and ClipShots but not BBC. Class-balanced calibration error also shows why very small aggregate ECE values should not be interpreted as uniformly calibrated boundary probabilities.

The revised implementation records exact cache identities, split manifests, frozen selection decisions, and artifact checksums. Completing the scheduled three-seed study requires restoring the original AutoShot base checkpoint; until then, optimization variability remains unresolved. Further work should also reserve an independent calibration split and address cross-domain precision on ClipShots.

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